

Statistics in Health Sciences - Assignment 1

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1 Introduction

The aim of this assignment is to apply the framework proposed by Basagaña et al. to the data set `copd` in the file `copd redux.RData`. Data `copd` is a subset of 45 variables of the original data analyzed in the cited paper.

2 Analysis

2.1 Description of missingness

Provide the same missingness description than in the original analysis. Specifically, reproduce the following text that is shown in the original paper:

“The subset used here included 342 participants and 85 variables. The range of missing data in the different variables went from 0% to 47.7%, with 68.2% of the variables having less than 5% of the data missing. Only 13.7% of the participants had complete data on all 85 variables. Overall, 5.9% of the information was missing; that is, 5.9% of the total of 29,070 cells ($85 \times 342 = 29,070$) had missing data.”

Instructions:

- Note that, since you are working with a subset of the original data set, red numbers above are not expected to be the same than in your results.
- In your `Rnw` document, the text above must be able to automatically update all red numbers in the case of changing the values of `copd`. I.e., if data change, then all red numbers must also automatically update accordingly in your `pdf` when compiling your `Rnw`. As a hint, you can follow this procedure:
 1. For each red number, create an R object, which will depend on data and gives the desired quantity (i.e., if data change, the value in the object changes accordingly),
 2. To force rounding a quantity to a given decimal precision, you can use `sprintf()` (as seen in Exercise 0). Do not use `round()`,
 3. To get the thousands comma separator as in “29,070”, you can use `prettyNum()`,
 4. Once you have an R object for each red number formatted as desired, use `\Sexpr{}` to insert each of them inline in $\text{\LaTeX}$ (as seen in Exercise 0).

```
> n_obs <- nrow(copd) # Number of participants
> n_vars <- ncol(copd) # Number of variables
> nans <- 100*colMeans(is.na(copd)) # Percentage of missing data by variable
> min_nan <- sprintf("%.2g", min(nans)) # MIRAR # Lower percentage
> max_nan <- sprintf("%.2g", max(nans)) # Upper percentage
>
> # Variables with less than 5% of data missing (percentage)
> less_5 <- sprintf("%.2g", 100*length(nans[nans<5])/n_vars)
>
> # Percentage of complete cases
> comp_cases <- sprintf("%.2g", 100*mean(complete.cases(copd)))
>
> # Percentage of missing data
> miss_total <- sprintf("%.2g", 100*mean(is.na(copd)))
>
> # Total cells observed (variables*observations)
> total_cells = prettyNum(n_obs*n_vars, big.mark = ",", decimal.mark = ".")
```

“The subset used here included 342 participants and 45 variables. The range of missing data in the different variables went from 0% to 32%, with 69% of the variables having less than 5% of the data missing. Only 33% of the participants had complete data on all 45 variables. Overall, 5.4% of the information was missing; that is, 5.4% of the total of 15,390 cells ($45 \times 342 = 15,390$) had missing data.”

2.2 Multiple imputation

Apply multiple imputation to copd data using 100 imputations. Explain in a reasoned manner how you decided to set the values of the arguments `method` and `predictorMatrix` when using the `mice()` function. Show your code.

```
> # Display of default imputation methods for each variable
> ini <- mice(copd, maxit=0)
> ini

## Class: mids
## Number of multiple imputations: 5
## Imputation methods:
##           id index_barthel2v1 quo_calcpstv1 rv_prepv1 incp_fevv1 fevp_postv1
##           ""                  "pmm"          ""          "pmm"          "pmm"
## fevp_postv1 rv_tlcvt1 ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_prepv1
##           ""                  "pmm"          "pmm"          "pmm"          "pmm"
## sin11_v1 vas2_v1 ic_prepv1 iam_chv1 sin15_v1
## "logreg" "pmm" "pmm" "logreg" "logreg"
## cil8_s_mv1 sin7_v1 sin8_v1 ch_v1 vc_prepv1
## "pmm" "logreg" "logreg" "pmm" "pmm"
## dtdve_ecov1 cccs1_v0 kcals_set_v0 impactsv1 monoA_v1
## "pmm" "logreg" "pmm" "pmm" "pmm"
## lvmass_ase_ecov1 vtit_catv1 urg_ambv0 po2_v1 ahpv_v1
## "pmm" "logreg" "" "pmm" "pmm"
## pro_v1 basA_v1 fc_pmv1 tad_v1 col_v1
## "pmm" "pmm" "pmm" "pmm" "pmm"
## sin5_v1 bc_v1 fvcv_postv1 vas1_v1 sin6_v1
## "logreg" "pmm" "" "pmm" "logreg"
## graudisn_v1 cpcr_mg_l_mediav1 neutA_v1 ahpl_v1 activityv1
## "polyreg" "pmm" "pmm" "pmm" "pmm"

## PredictorMatrix:
##           id index_barthel2v1 quo_calcpstv1 rv_prepv1 incp_fevv1 fevp_postv1
## id 0 1 1 1 1 1
## index_barthel2v1 1 0 1 1 1 1
## quo_calcpstv1 1 1 0 1 1 1
## rv_prepv1 1 1 1 0 1 1
## incp_fevv1 1 1 1 1 0 1
## fevp_postv1 1 1 1 1 1 0
##           rv_tlcvt1 ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_prepv1
## id 1 1 1 1 1 1 1
## index_barthel2v1 1 1 1 1 1 1 1
## quo_calcpstv1 1 1 1 1 1 1 1
## rv_prepv1 1 1 1 1 1 1 1
## incp_fevv1 1 1 1 1 1 1 1
## fevp_postv1 1 1 1 1 1 1 1
##           iam_chv1 sin15_v1 cil8_s_mv1 sin7_v1 sin8_v1 ch_v1 vc_prepv1 dtdve_ecov1
## id 1 1 1 1 1 1 1 1
## index_barthel2v1 1 1 1 1 1 1 1 1
## quo_calcpstv1 1 1 1 1 1 1 1 1
## rv_prepv1 1 1 1 1 1 1 1 1
## incp_fevv1 1 1 1 1 1 1 1 1
## fevp_postv1 1 1 1 1 1 1 1 1
##           cccs1_v0 kcals_set_v0 impactsv1 monoA_v1 lvmass_ase_ecov1 vtit_catv1
## id 1 1 1 1 1 1
## index_barthel2v1 1 1 1 1 1 1
## quo_calcpstv1 1 1 1 1 1 1
## rv_prepv1 1 1 1 1 1 1
## incp_fevv1 1 1 1 1 1 1
## fevp_postv1 1 1 1 1 1 1
##           urg_ambv0 po2_v1 ahpv_v1 pro_v1 basA_v1 fc_pmv1 tad_v1 col_v1 sin5_v1
## id 1 1 1 1 1 1 1 1
## index_barthel2v1 1 1 1 1 1 1 1 1
## quo_calcpstv1 1 1 1 1 1 1 1 1
## rv_prepv1 1 1 1 1 1 1 1 1
## incp_fevv1 1 1 1 1 1 1 1 1
## fevp_postv1 1 1 1 1 1 1 1 1
##           bc_v1 fvcv_postv1 vas1_v1 sin6_v1 graudisn_v1 cpcr_mg_l_mediav1 neutA_v1
## id 1 1 1 1 1 1 1
## index_barthel2v1 1 1 1 1 1 1 1
```

```
## quo_calcpstv1      1      1      1      1      1      1      1
## rv_prepv1          1      1      1      1      1      1      1
## incp_fevv1         1      1      1      1      1      1      1
## fevp_postv1        1      1      1      1      1      1      1
##                   ahpl_v1 activityv1
## id                  1      1
## index_barthel2v1    1      1
## quo_calcpstv1        1      1
## rv_prepv1            1      1
## incp_fevv1           1      1
## fevp_postv1          1      1
```

We can see that the mice package suggest imputing initially with de mean for continuous variables and with a logistic regression for binary ones. Further research on the dataset has shown that the variables have no deterministic relation. So we do not change the imputation methods. We have a variable (`id`) that don't have any role in the imputation process, so we change `predictorMatrix` and `methods` according to that.

```
> # Change in parameters
> meth <- ini$method
> pred <- ini$predictorMatrix
>
> meth["id"] <- "" # do not impute "id"
> pred["id", ] <- 0 # we don't want "id" to be imputed
> pred[, "id"] <- 0 # we don't want "id" to be predictor
```

```
> # Imputation of 100 datasets
> M <- 2 # Number of imputations for each missing value
> imp <- mice(data = copd,
+             m = M,
+             method = meth,
+             predictorMatrix = pred,
+             maxit = 20,
+             printFlag = FALSE,
+             seed = 666)
>
> imp$imp

## $id
## [1] 1 2
## <0 rows> (o 0- extensión row.names)
##
## $index_barthel2v1
##      1  2
## 31 100 100
## 64 100 100
## 187 100 100
## 198  80 100
## 240  90 100
## 241 100 100
## 242 100 100
## 245 100 100
## 302  90 100
##
## $quo_calcpstv1
## [1] 1 2
## <0 rows> (o 0- extensión row.names)
##
## $rv_prepv1
##      1      2
## 31 190.00200 156.37891
## 73 175.15790 140.45828
## 84 179.54126 166.25749
## 85 174.01822 154.18134
## 87 147.11288 142.57919
## 91 266.79022 199.57353
## 96 194.25191 152.39517
```

```

## 99 134.39871 114.54136
## 105 175.47922 154.18134
## 109 304.11612 304.11612
## 110 130.83453 123.91756
## 115 153.99147 173.26074
## 117 233.97670 218.04686
## 121 130.73349 152.96429
## 123 228.64987 94.60959
## 130 114.64743 156.37891
## 139 156.37891 144.07747
## 146 168.70444 128.22917
## 159 87.62526 201.72061
## 172 114.49117 168.70444
## 180 119.85428 135.60764
## 188 137.62262 115.28805
## 189 123.91756 115.28805
## 208 233.97670 128.13847
## 218 147.11288 152.96429
## 277 148.80103 212.59613
## 301 110.02028 177.64473
##
## $incp_fevv1
##      1      2
## 62  4.999995 32.0610771
## 82  7.526881 17.7914085
## 87 11.023622 21.5189819
## 91 -1.408460 -0.4854364
## 126 9.223304 -3.9682503
## 144 6.766911 1.6393427
## 153 11.023622 3.0769293
## 160 0.000000 5.9171538
## 161 2.127658 4.3103509
## 170 -4.186053 -8.0952368
## 179 20.430107 11.0236216
## 186 -4.186053 6.7669106
## 311 -3.968250 4.5454569
## 332 4.237294 4.3902516
##
## $fevp_postv1
## [1] 1 2
## <0 rows> (o 0- extensión row.names)
##
## $rv_tlc1
##      1      2
## 31 54.76575 50.40129
## 73 43.56725 50.40129
## 84 71.79082 64.67818
## 85 56.49350 56.31769
## 87 56.16083 46.26866
## 91 71.79082 72.55872
## 96 69.04762 72.95850
## 99 55.23521 62.98450
## 105 67.72152 64.48781
## 109 72.83621 75.81699
## 110 39.49580 39.85185
## 115 52.24625 63.09720
## 117 72.79793 63.47223
## 121 62.16617 61.83007
## 123 62.39814 40.65245
## 130 56.31769 58.24468
## 139 45.96651 53.05515
## 146 55.53359 49.01640
## 159 43.56725 62.77602
## 172 60.17965 64.69428
## 180 45.03043 64.28571
## 188 50.07215 46.30872
## 189 49.11937 35.66553
## 208 63.55263 41.05264
## 218 53.71747 51.84275
## 277 59.30599 54.39560

```

```

## 301 45.68289 40.78550
##
## $ctnf_s_mv1
##      1      2
## 31  0.04500 3.71560
## 54  3.82460 0.04500
## 91  0.32391 3.52540
## 98  0.04500 0.04500
## 187 0.04500 0.56940
## 188 0.04500 3.82460
## 198 0.34994 0.04500
## 215 2.28680 0.35454
## 233 0.04500 0.04500
## 234 1.10120 1.55040
## 238 0.04500 0.04500
## 254 1.04920 3.71560
##
## $vtdvdbid_ecov1
##      1  2
##  3   37 31
##  5   27 31
##  6   33 27
##  7   34 29
##  8   37 29
##  9   34 35
## 10   35 23
## 11   35 40
## 12   35 31
## 14   31 33
## 15   31 29
## 19   33 32
## 21   29 28
## 22   29 31
## 23   39 33
## 25   33 34
## 31   26 28
## 38   32 27
## 39   44 44
## 51   31 28
## 55   30 40
## 56   31 29
## 57   37 35
## 59   30 26
## 82   35 31
## 84   27 27
## 85   33 38
## 91   28 31
## 94   31 34
## 96   34 31
## 98   28 33
## 104  26 34
## 106  29 31
## 114  37 32
## 118  33 30
## 134  40 37
## 137  35 30
## 140  38 27
## 142  38 29
## 145  32 34
## 150  28 31
## 154  32 30
## 173  33 38
## 178  28 29
## 187  27 30
## 188  35 35
## 189  37 33
## 190  40 35
## 191  29 32
## 192  33 34
## 193  28 26

```

```

## 194 31 31
## 195 31 27
## 196 44 31
## 197 33 29
## 198 23 32
## 199 27 32
## 200 31 31
## 201 33 44
## 202 27 38
## 203 33 37
## 204 27 37
## 205 30 40
## 206 40 31
## 207 32 34
## 208 40 40
## 209 26 29
## 210 32 33
## 212 35 27
## 213 29 37
## 215 26 30
## 217 35 31
## 219 30 27
## 220 37 32
## 221 28 31
## 222 28 31
## 224 35 29
## 229 32 32
## 230 32 33
## 232 28 27
## 233 29 26
## 234 29 28
## 235 32 32
## 236 34 31
## 240 35 30
## 241 32 23
## 243 27 27
## 244 30 35
## 245 37 29
## 246 31 32
## 249 28 32
## 251 28 33
## 255 37 37
## 271 31 35
## 272 30 29
## 278 32 29
## 279 27 26
## 280 27 32
## 287 37 35
## 288 35 33
## 292 30 28
## 303 33 34
## 304 32 33
## 305 27 29
## 308 25 29
## 313 35 32
## 318 27 28
## 335 27 27
##
## $pp_ecov1
##      1      2
## 31  11.0 11.0
## 38  11.0 11.0
## 39  13.0 11.0
## 48  10.0 13.0
## 51   9.0 10.0
## 53  11.0 11.0
## 65   9.0  9.0
## 82   8.0  9.0
## 91  11.0  9.0
## 114 11.0  9.0

```

```

## 134 11.0 9.0
## 137 11.0 11.0
## 144 13.0 10.0
## 145 9.0 9.0
## 148 8.0 8.0
## 150 9.0 10.0
## 156 8.0 9.0
## 173 10.0 11.0
## 187 10.0 10.0
## 188 9.0 10.0
## 193 9.0 9.0
## 198 10.0 9.0
## 203 13.0 10.0
## 205 11.0 11.0
## 210 9.0 11.0
## 212 9.0 8.0
## 213 9.0 10.0
## 215 10.0 10.0
## 217 10.0 8.0
## 219 11.0 12.0
## 220 12.0 10.0
## 221 11.0 10.0
## 222 11.0 10.0
## 224 9.0 10.0
## 229 11.5 9.0
## 230 10.0 11.0
## 232 12.0 10.0
## 233 10.0 9.0
## 234 12.0 11.0
## 235 9.0 10.0
## 236 10.0 9.0
## 240 9.0 9.0
## 241 9.0 9.0
## 243 13.0 11.5
## 244 10.0 10.0
## 277 9.0 8.0
## 278 10.0 8.0
## 288 11.0 9.0
## 292 10.0 10.0
## 301 9.0 10.0
## 304 10.0 10.0
## 320 9.0 11.0
## 325 10.0 11.0
## 335 11.0 10.0
##
## $sin11_v1
##      1 2
## 31  1 1
## 64  0 0
## 187 0 0
## 198 0 1
##
## $vas2_v1
##      1 2
## 31  8.5 4.0
## 64  5.0 5.0
## 187 6.0 4.4
## 198 7.0 4.0
##
## $ic_prepv1
##      1      2
## 73  77.49661 78.91282
## 84  54.70592 65.09283
## 85  89.25582 68.98782
## 87 104.88959 97.75817
## 91  32.85936 42.27396
## 99  42.70299 62.30412
## 105 47.36942 35.50232
## 109 29.84534 35.55212
## 115 78.91282 71.43972

```

```

## 121 35.50232 45.54313
## 123 77.49661 87.21473
## 130 57.39879 56.28346
## 146 68.49036 79.16180
## 159 91.69626 54.52460
## 172 59.21489 42.70299
## 180 42.70299 59.21489
## 188 59.35907 69.05781
## 189 103.98996 100.07884
## 218 92.72242 79.99995
##
## $iam_chv1
##      1 2
## 188 1 1
## 198 0 1
## 278 0 0
##
## $sin15_v1
##      1 2
## 31  1 0
## 64  0 0
## 187 0 0
## 198 0 1
##
## $cil8_s_mv1
##      1      2
## 31  9.92210 4.92940
## 54 11.41900 23.49700
## 56  4.30155 2.60710
## 91  1.54125 3.92915
## 98  8.87310 3.58110
## 187 4.64540 3.96030
## 188 16.57900 6.50330
## 198 4.50310 2.50265
## 208 4.56365 5.86350
## 215 3.27120 3.67125
## 233 3.59100 2.90325
## 234 4.56075 4.28700
## 238 1.54125 3.70075
## 254 5.72960 2.48030
##
## $sin7_v1
##      1 2
## 31  1 0
## 64  1 0
## 187 0 0
## 198 0 1
##
## $sin8_v1
##      1 2
## 31  1 1
## 64  1 1
## 187 0 0
## 198 0 1
##
## $ch_v1
##      1 2
## 95  0.6 4.8
## 109 1.1 2.6
## 114 4.8 0.9
## 118 1.1 0.9
## 119 1.8 7.8
## 120 0.7 0.5
## 121 2.1 2.6
## 124 4.1 2.6
## 131 2.6 0.9
## 197 0.2 0.3
## 201 0.6 3.9
## 249 0.9 0.7
## 250 4.1 3.9

```

```

## 259 1.2 0.6
## 269 4.1 0.9
## 278 1.7 0.5
## 307 0.6 2.3
## 342 0.7 0.9
##
## $vc_prepv1
##      1      2
## 84  43.34512 45.59875
## 85  73.80546 74.88763
## 87  82.23821 98.87781
## 91  48.84620 43.34512
## 96  55.31234 36.51057
## 99  65.28603 68.47959
## 105 48.84620 46.14307
## 109 46.05139 43.67628
## 115 67.39462 65.40956
## 121 53.32501 49.36518
## 123 81.78346 98.87781
## 130 67.87232 57.75216
## 146 72.69066 73.65106
## 159 85.63545 63.95637
## 172 63.57674 58.62754
## 180 68.74204 61.92160
## 188 106.87608 79.62598
## 189 97.70652 113.37632
## 218 69.73008 73.73167
##
## $dtdve_ecov1
##      1  2
## 31  60 54
## 38  40 56
## 39  48 40
## 48  48 56
## 51  56 46
## 65  43 62
## 82  64 57
## 91  40 49
## 114 51 43
## 134 59 46
## 137 57 40
## 144 63 53
## 145 58 54
## 148 57 48
## 150 48 47
## 156 49 44
## 173 48 51
## 187 60 59
## 188 46 53
## 193 49 53
## 198 58 41
## 203 40 46
## 205 54 55
## 210 49 56
## 212 42 66
## 213 44 44
## 215 47 60
## 217 50 53
## 219 56 59
## 220 49 52
## 221 53 49
## 222 56 54
## 224 48 56
## 229 49 50
## 230 38 50
## 232 46 49
## 233 40 45
## 234 56 57
## 235 53 45
## 236 60 54

```

```

## 240 63 48
## 241 48 56
## 243 60 50
## 244 53 56
## 278 47 54
## 288 47 51
## 292 49 56
## 304 47 55
## 335 40 58
##
## $cccs1_v0
##      1 2
## 87  1 0
## 89  0 0
## 93  0 1
## 245 0 0
## 316 0 0
##
## $kcalcs_set_v0
##      1 2
## 187 4062 1584
##
## $impactsv1
##      1 2
## 31  41.349514 43.638657
## 64  21.139862 12.276891
## 187 15.506658  3.552047
## 198  7.243366 30.767778
##
## $monoA_v1
##      1 2
## 31  0.4602000 0.60800
## 187 0.5856000 0.23898
## 188 0.7777799 0.54270
## 198 0.3801600 0.39558
## 254 0.3696000 1.50420
## 298 0.4453600 0.13656
## 301 0.4750800 0.50160
##
## $lvmass_ase_ecov1
##      1 2
## 31  253933.7 287863.4
## 38  152816.5 287863.4
## 39  233723.5 182432.4
## 48  175436.1 302913.5
## 51  175436.1 152351.5
## 53  244200.1 181376.6
## 65  145041.5 255109.3
## 82  320314.8 187416.9
## 91  137120.9 158213.7
## 114 188667.4 131506.5
## 134 255970.4 188667.4
## 137 248715.4 156334.2
## 144 348881.5 248715.4
## 145 213297.1 199797.1
## 148 233723.5 147231.3
## 150 100833.2 187416.9
## 156 163381.3 163381.3
## 173 152351.5 239263.0
## 187 287863.4 181307.5
## 188 142107.9 205770.0
## 193 180796.7 187492.6
## 198 213297.1 131506.5
## 203 244200.1 162954.5
## 205 255109.3 191034.5
## 210 158213.7 179362.3
## 212  96759.7 255970.4
## 213 137216.4 151954.6
## 215 163854.7 240747.3
## 217 211401.8 147184.7

```

```

## 219 263817.8 348881.5
## 220 213297.1 213281.3
## 221 213297.1 169595.5
## 222 240747.3 181307.5
## 224 163381.3 227134.9
## 229 279826.3 144622.2
## 230 188667.4 232494.7
## 232 199797.1 187416.9
## 233 151954.6 158222.9
## 234 263817.8 234014.8
## 235 199898.6 163381.3
## 236 248715.4 227134.9
## 240 219990.5 175436.1
## 241 182432.4 227134.9
## 243 412904.7 199797.1
## 244 244200.1 263817.8
## 277 239263.0 202895.5
## 278 211401.8 226804.6
## 288 200124.1 152351.5
## 292 147231.3 174419.4
## 301 145041.5 234014.8
## 304 169251.9 226804.6
## 320 179532.9 240747.3
## 325 227134.9 199898.6
## 335 157608.0 247000.6
##
## $vtit_catv1
##      1 2
## 2    0 0
## 6    0 1
## 7    1 0
## 10   1 1
## 19   1 0
## 21   1 0
## 22   1 1
## 23   1 1
## 25   1 0
## 31   1 0
## 38   0 1
## 39   1 1
## 51   1 1
## 55   0 1
## 56   0 1
## 57   1 1
## 76   1 0
## 82   0 1
## 87   0 1
## 91   1 1
## 96   0 1
## 98   0 1
## 101  1 0
## 104  0 0
## 114  1 0
## 134  1 0
## 137  1 1
## 139  1 1
## 145  0 0
## 150  0 1
## 154  1 0
## 155  1 1
## 157  1 0
## 173  1 0
## 178  0 1
## 187  0 1
## 188  0 0
## 193  0 1
## 198  0 0
## 201  0 1
## 203  1 0
## 205  0 0

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## 207 1 1
## 210 1 1
## 212 1 0
## 213 0 1
## 215 1 1
## 217 1 1
## 219 0 1
## 220 0 0
## 221 0 1
## 222 1 1
## 224 0 1
## 229 1 0
## 230 0 1
## 232 1 1
## 233 1 0
## 234 1 0
## 235 1 1
## 236 1 0
## 240 0 1
## 241 1 1
## 242 0 0
## 243 1 0
## 244 1 1
## 251 1 0
## 255 1 1
## 257 0 0
## 259 0 0
## 262 1 1
## 263 1 0
## 277 0 0
## 278 0 0
## 280 1 1
## 292 0 1
## 301 0 0
## 302 1 1
## 303 0 0
## 304 1 0
## 312 0 0
## 313 0 1
## 326 0 0
## 327 1 1
## 328 0 1
## 329 0 0
## 330 0 0
## 335 1 1
##
## $urg_ambv0
## [1] 1 2
## <0 rows> (o 0- extensión row.names)
##
## $po2_v1
##      1      2
## 92 75.0 68.2
## 95 71.2 65.3
## 110 90.0 92.0
## 114 57.3 81.1
## 116 71.2 75.0
## 117 66.9 57.3
## 185 71.5 77.0
## 197 78.6 118.0
## 201 66.9 64.7
## 278 66.0 93.0
## 342 77.8 74.7
##
## $ahpf_v1
##      1      2
## 9      3.3 2.2
## 15     1.2 3.1
## 18     1.0 1.1
## 20     5.6 1.9

```

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## 26 2.8 1.7
## 31 2.1 3.8
## 41 2.6 4.5
## 51 2.5 5.5
## 57 3.6 2.5
## 82 2.3 3.6
## 104 2.5 0.7
## 105 6.0 2.7
## 115 1.2 2.1
## 134 1.5 2.8
## 135 5.6 2.3
## 136 2.6 2.2
## 139 1.0 3.1
## 140 3.1 1.9
## 141 0.8 2.2
## 142 4.2 4.2
## 143 4.0 1.8
## 144 3.0 2.2
## 145 2.2 2.8
## 163 5.1 4.0
## 172 3.2 2.2
## 177 2.6 3.2
## 178 3.6 3.4
## 179 1.0 1.5
## 180 2.2 2.9
## 182 3.8 2.8
## 185 2.8 2.3
## 186 2.2 3.4
## 187 3.0 3.2
## 188 1.2 1.5
## 193 2.6 2.6
## 196 6.0 3.1
## 205 2.8 1.1
## 209 4.5 3.3
## 212 5.5 1.8
## 214 2.0 3.3
## 215 0.8 2.2
## 223 2.3 3.2
## 240 1.0 2.4
## 241 3.6 2.5
## 247 4.5 3.5
## 252 3.5 4.1
## 258 5.3 2.5
## 263 4.1 3.8
## 269 2.9 2.6
## 274 1.9 2.2
## 278 2.8 4.0
## 304 2.2 3.0
## 309 1.2 1.4
## 311 5.5 2.3
## 312 2.6 2.8
## 336 3.0 3.0
##
## $pro_v1
##      1      2
## 31 7.85 6.70
## 98 7.70 7.70
## 187 6.90 7.50
## 188 8.20 7.22
## 189 7.30 7.20
## 190 7.17 7.80
## 191 6.52 7.50
## 194 7.17 7.50
## 198 6.84 6.70
## 241 6.90 7.37
## 245 7.72 9.10
## 301 7.72 7.56
##
## $basA_v1
##      1      2

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## 31 0.01650 0.03600
## 187 0.00810 0.05460
## 188 0.05340 0.02800
## 198 0.03396 0.12090
## 254 0.04150 0.05680
## 298 0.04080 0.06016
## 301 0.03104 0.04528
##
## $fc_pmv1
##      1      2
## 17  2  5.0
## 18  4  1.0
## 26  0  3.0
## 29  5  4.0
## 31  5  4.0
## 35  5  7.0
## 36  2  5.0
## 46  0  1.0
## 48  5  0.0
## 63  0  2.0
## 65  8  5.0
## 67  1  0.0
## 68  2  0.0
## 69  5  0.0
## 80  1  3.0
## 87  5  4.0
## 91  2  7.0
## 92  0  1.0
## 96  1  0.0
## 115 9  5.0
## 139 8  4.0
## 178 8  2.0
## 187 7  3.0
## 188 8  0.0
## 198 4  0.0
## 203 5  0.0
## 211 0  0.0
## 212 0  1.0
## 214 3  0.0
## 215 0  0.0
## 252 5  5.2
## 278 0  0.0
## 281 0  1.0
## 282 3  5.0
## 292 4  5.0
## 301 8  3.0
## 302 0  7.0
## 304 4  2.0
## 320 5 10.0
##
## $tad_v1
##      1      2
## 95  77 111
## 111 90  75
## 188 70  63
## 198 90  50
## 278 70  80
##
## $col_v1
##      1      2
## 3  133.6 186.0
## 31 257.0 127.0
## 169 203.0 285.0
## 170 192.0 222.0
## 171 194.0 244.0
## 187 261.0 211.0
## 188 124.0 127.2
## 197 267.0 175.0
## 198 168.0 241.0
## 301 250.0 252.4

```

```

##
## $sin5_v1
##      1 2
## 31  0 0
## 64  1 1
## 187 1 0
## 198 0 0
##
## $bc_v1
##      1 2
## 17  0.6 0.0
## 18  0.0 0.0
## 26  0.0 2.0
## 29  1.0 0.0
## 31  3.0 0.0
## 35  0.0 6.0
## 36  0.0 0.0
## 46  0.0 0.0
## 48  1.0 0.0
## 63  0.0 1.0
## 65  3.0 5.0
## 67  0.0 0.0
## 68  0.0 0.0
## 69  1.0 1.0
## 80  0.0 0.0
## 87  6.0 5.0
## 91  0.0 0.5
## 92  0.0 0.0
## 115 2.0 0.0
## 139 0.0 0.0
## 178 2.0 1.0
## 187 6.0 3.0
## 188 1.0 0.0
## 198 6.0 0.0
## 203 0.0 0.0
## 211 0.0 1.0
## 212 2.0 0.0
## 214 3.0 0.0
## 215 0.0 0.0
## 252 4.0 5.0
## 278 0.0 0.0
## 281 5.0 0.0
## 292 6.0 4.0
## 301 0.0 0.5
## 302 4.0 0.0
## 304 1.0 0.0
## 320 0.0 2.0
##
## $fvcp_postv1
## [1] 1 2
## <0 rows> (o 0- extensión row.names)
##
## $vas1_v1
##      1 2
## 31  9.0 2.7
## 64  2.0 1.9
## 187 5.7 0.3
## 198 2.0 2.0
##
## $sin6_v1
##      1 2
## 31  1 0
## 64  1 0
## 187 1 1
## 198 0 1
##
## $graudisn_v1
##      1 2
## 31  3 1
## 64  2 0

```

```

## 187 1 2
## 198 4 2
##
## $cpcr_mg_l_mediav1
##      1      2
## 31  0.0800 0.0800
## 54  0.0800 0.3420
## 56  3.1050 0.3420
## 63  0.9580 1.5850
## 85  0.5050 0.5375
## 91  3.0050 1.6200
## 98  1.7200 0.5725
## 104 0.3420 0.1970
## 187 0.9480 0.0800
## 188 2.9650 0.4080
## 198 0.2755 0.3910
## 215 0.2765 0.9580
## 233 0.3690 0.3515
## 234 0.7865 1.0200
## 238 0.5095 0.6538
## 254 0.3035 1.3800
## 303 0.0800 0.4180
##
## $neutA_v1
##      1      2
## 31  5.18445 5.18445
## 187 3.78190 3.38140
## 188 9.51400 5.68974
## 198 5.22416 6.58940
## 254 3.09570 4.50850
## 298 2.03943 3.38140
## 301 3.00294 4.20480
##
## $ahpl_v1
##      1      2
## 9   1.0 1.0
## 10  3.3 1.8
## 14  2.3 4.0
## 15  1.2 2.6
## 18  1.6 0.5
## 20  5.0 2.5
## 22  0.6 3.2
## 26  0.5 0.8
## 31  0.4 3.0
## 51  5.0 5.0
## 57  3.2 4.0
## 68  1.2 1.2
## 69  0.5 5.2
## 72  2.5 2.1
## 82  1.2 2.1
## 86  3.0 4.0
## 92  1.0 3.8
## 104 4.0 1.9
## 107 4.1 3.0
## 108 3.4 2.1
## 112 1.4 3.2
## 113 4.0 1.8
## 115 3.3 1.0
## 134 2.0 3.2
## 135 2.2 1.6
## 136 2.3 2.2
## 137 3.3 2.7
## 139 1.0 2.3
## 140 4.0 2.7
## 141 1.3 4.1
## 142 5.0 3.2
## 143 3.0 2.0
## 144 4.8 5.0
## 145 0.6 0.8
## 147 1.8 2.1

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```
## 164 2.0 5.2
## 168 2.0 2.7
## 175 2.5 4.0
## 176 4.0 4.0
## 179 0.4 2.0
## 180 3.0 1.4
## 185 4.0 2.7
## 187 5.2 2.7
## 188 0.5 2.2
## 190 2.9 2.0
## 191 2.7 5.5
## 193 3.0 2.1
## 196 1.8 5.2
## 200 1.1 1.8
## 205 0.6 2.5
## 209 4.6 1.8
## 212 4.0 2.2
## 215 0.4 0.9
## 217 2.0 1.8
## 240 0.5 2.1
## 241 6.0 4.0
## 252 2.4 4.0
## 263 4.6 3.6
## 274 0.7 3.0
## 278 1.0 3.3
## 304 2.5 1.6
## 309 2.5 2.7
## 311 4.1 2.0
## 312 1.9 2.0
## 313 4.0 2.5
## 336 3.3 2.5
##
## $activityv1
##           1           2
## 31  80.53098 73.03780
## 64  23.34794 17.13671
## 187 47.68836 23.33140
## 198 72.88892 54.53643
```